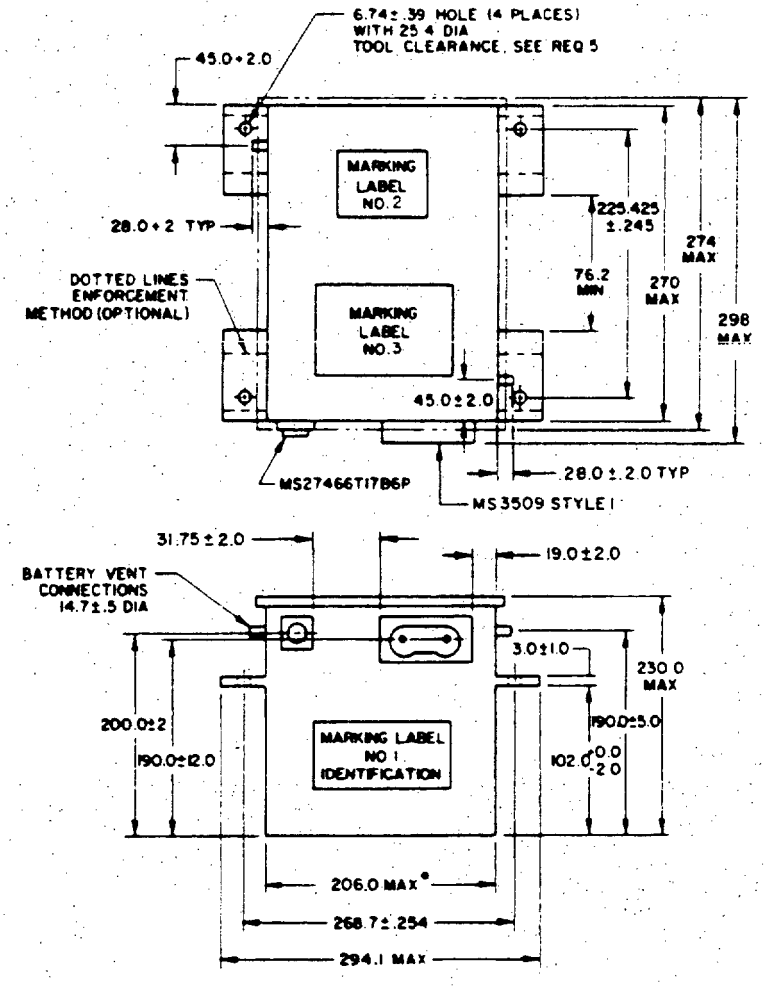
MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 24-AMPERE-HOUR

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the storage battery described herein shall consist of this document and the latest issue of specification MIL-B-8565.



NOTE

1. ALL DIMENSIONS ARE MILLIMETERS.
*ALLOW ADDITIONAL 4MM WIDTH FOR COVER.

FIGURE 1. Dimensions P/N D8565/7-1.

AMSC N/A FSC 6140

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COORDINATES OF CURVES STRAIGHT LINE BETWEEN COORDINATES		
CURRENT IN AMPS	POTENTIAL IN VOLTS	START TIME IN SECONDS
280	17.5	0
175	19.0	2
130	20.0	4
100	20.5	8
80	21.0	12
60	21.5	50
40	22.0	60

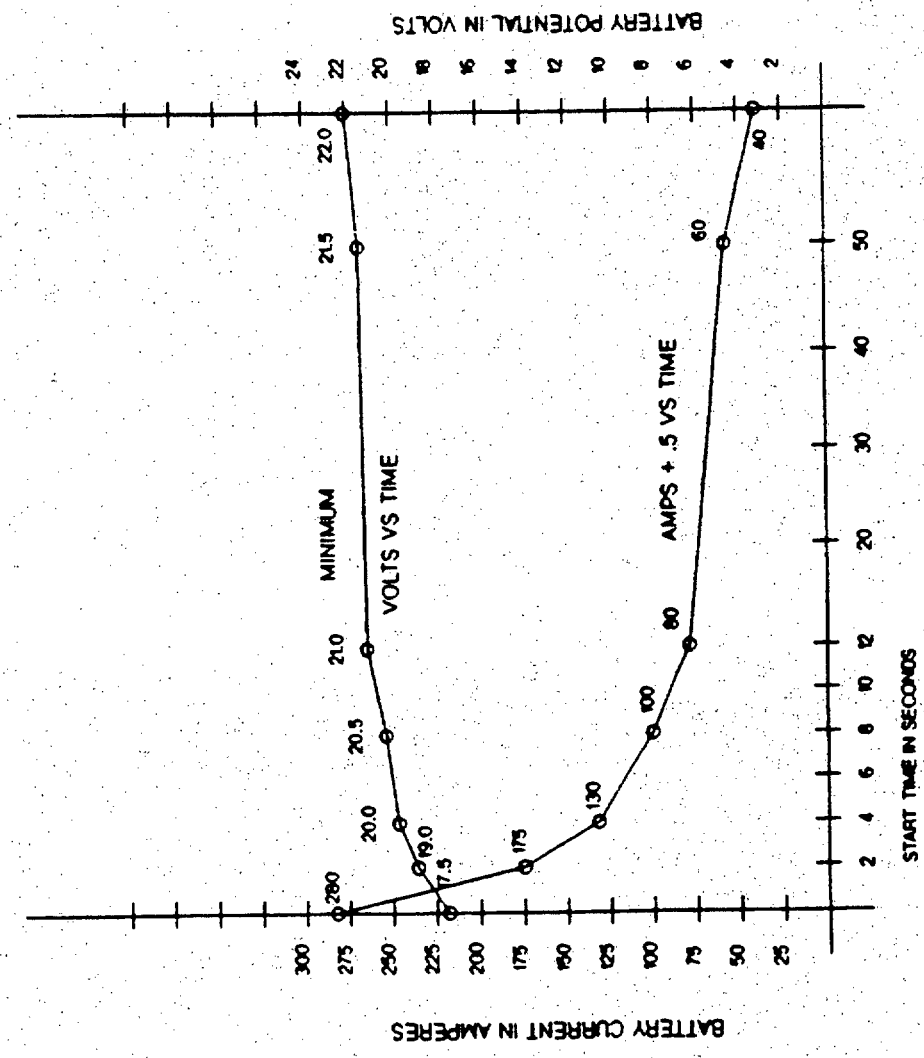
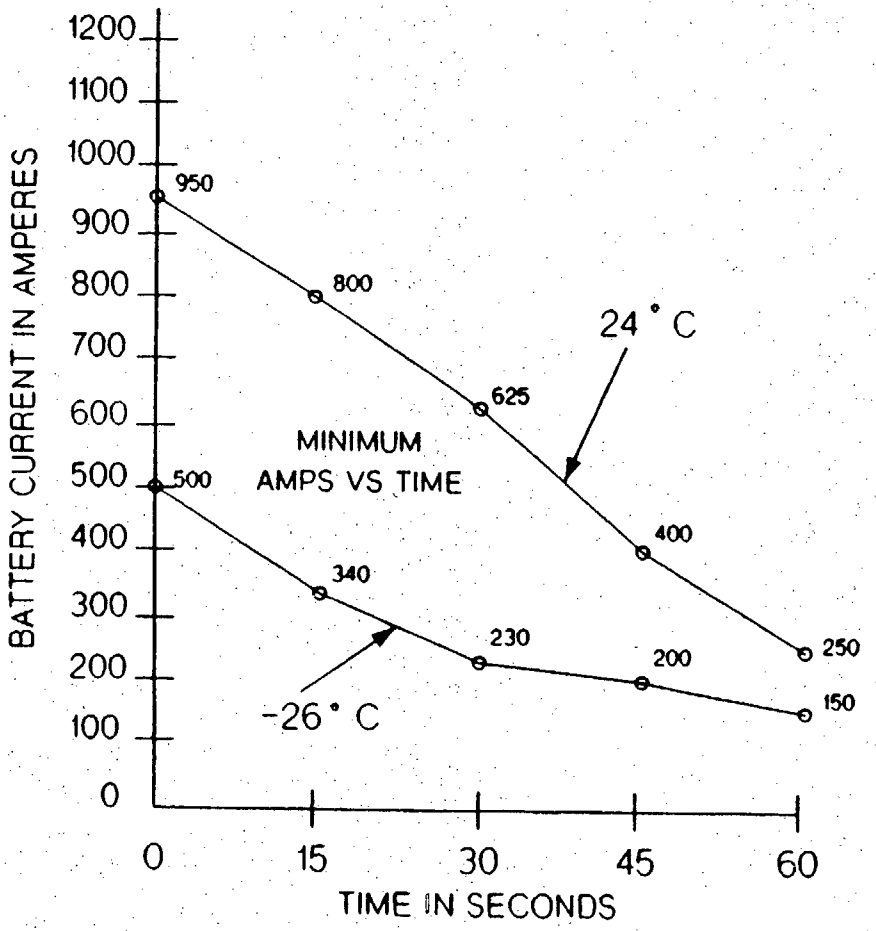


FIGURE 2. Start cycle requirements

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COORDINATES OF CURVES STRAIGHT LINE BETWEEN COORDINATES		
TIME IN SECONDS	CURRENT IN AMPERES AT	
	24°C	-26°C
0*	950	500
15	800	340
30	625	230
45	400	200
60	250	150

FIGURE 3. Constant voltage (14.0 V) discharge curves (3.6.8 and 4.6.11).

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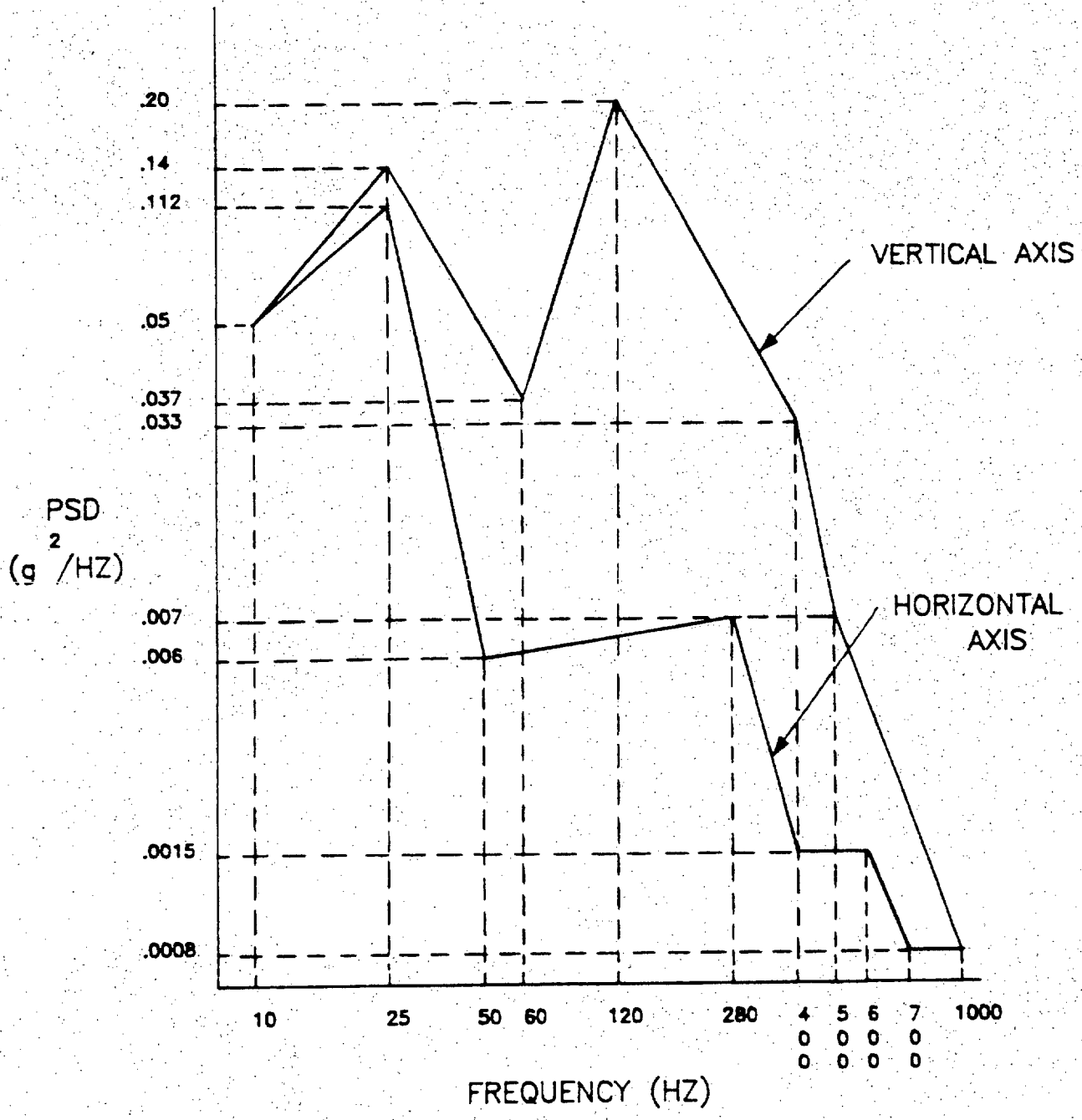


FIGURE 4. Random vibration test levels.

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REQUIREMENTS

1. The dimensions of the battery shall be as shown on Figure 1.

2. The part number of the battery shall be D8565/7-1.

3. The rated capacity of the battery shall be 24 ampere-hours (24.0 AH/1 HR/21°C/18.0 V at end of life). 1C = 24.0 amperes.

4. The weight of the battery shall not exceed 29.0 Kg. The weight of the batteries supplied under contract, subsequent to qualification, shall not vary by more than ± .6 Kg from the qualification sample.

5. Mounting holes shall have a coating-free surface, 12.7 mm diameter both sides.

6. Aircraft installation. For installation in the aircraft, the battery shall be supported as shown in Figure 1 of MIL-B-8565. Vibration isolation shall be provided, if necessary to insure that the vibration level does not exceed the requirements of the 2-year accelerated life test of 4.6.13. If vibration isolators are used, sufficient sway clearance between the battery and the mounting structure shall be provided. Battery life in the aircraft shall be 2 years.

7. The battery shall conform to military specification MIL-B-8565 except as modified by the following:

3.5.5 Receptacles. Add the following: "The receptacle for the battery's positive and negative connections shall be dimensionally equal to MS3509, Style 1."

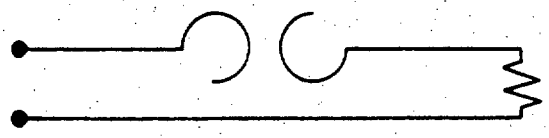
3.6.1.1 Add the following: "An auxiliary receptacle, MS27466T17B6P, shall be located as shown on Figure 1.

a. Pin A shall be high side AC (input heater power).

b. Pin B shall be AC ground (heater return)."

3.6.3.2.1 Polarity marking. Add the following: "Marking for label 3 shown on Figure 1 shall show the following with lettering height of .15 inch minimum.

PIN A 115 VAC



PIN B AC GROUND"

3.6.8 Capacity and electrical performance. Add the following: "The discharge current shall not drop below the ampere values shown in Figure 3 during the 14.0-volt constant voltage discharge. Inability of the sample to deliver the specified minimum discharge current for 1 minute shall constitute a test failure."

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Add the following paragraphs:

3.6.23 Standby power output performance.

3.6.23.1 The battery, when discharged into a 16-ohm resistive load, shall maintain a terminal potential of 24.0 volts or greater for the discharge time and temperature conditions shown below.

Temperature (°C)	Discharge Time (Hours)
-26	1.0
-18	2.0
0	4.0
25	7.0
49	7.0

3.6.23.2 The battery, after completing a discharge into a 16-ohm load for the time and temperature conditions shown above, shall successfully complete three start cycles, with the energy required in Figure 2.

3.6.24 Cold and hot temperature charging.

3.6.24.1 Half-hour charging. The battery, having 12.0Ah capacity at temperatures in the range of -18°C to 55°C, shall be capable of delivering 18.0Ah capacity after a 28.25 ± 0.5V charging source has been applied to the battery for 30 minutes.

3.6.24.2 One-hour charging. The battery, when discharged at temperatures in the range of -40°C to -19°C at 24.0A to 18.0V, shall be capable of delivering 18.0Ah capacity after a 28.25 ± 0.5V charging source has been applied to the battery for 1.0 hour. The battery shall not be damaged when a 28.25 ± 0.5V charging source is applied continuously to the battery at -40°C.

3.6.24.3 Heaters. Heaters may be incorporated into the battery to meet the cold temperature charging requirements. The heaters shall utilize 115VAC power in accordance with MIL-STD-704, accessible at auxiliary connector pins A and B. Heater current shall not exceed 6A at 115V. Heater power will not be provided during any discharge performed in accordance with this specification except during vibration testing. Heater power will be provided during all constant-potential charges.

3.6.24.3.1 The heater element shall be fabricated and installed in the battery assembly using insulated heaters and wire assemblies so as to minimize the possibility of sparking.

3.6.24.3.2 The battery shall contain temperature limiting devices as needed to prevent overheating when the heater circuit is energized. Under any condition of operation specified herein, the battery case temperature shall not exceed 75°C due to heater and battery operation.

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If the battery case temperature exceeds 49°C, a caution stating "CAUTION HOT SURFACE" shall be shown on marking labels 1 and 3. Both temperature conditions above apply at an ambient temperature of 25° ± 5°C. The heater control devices shall be compatible with the battery heater current requirements. If thermal switches are used, they shall be a snap-action type in accordance with MIL-S-24236. The switches shall be designed and installed to prevent ignition of an explosive mixture of hydrogen and oxygen. All components of the heater circuit shall be located inside the battery case. In the event of primary heater control failure, the battery shall not reach a damaging temperature.

3.19.1 Ground storage. Delete and add the following: "The battery shall be capable of storage at temperature conditions from -26°C to 50°C for up to 30 days and maintain a sufficient state of charge to successfully complete three start cycles per Figure 2."

4.6.15 Charge and discharge test at low temperature. Delete.

4.6.20 Special tests. Add the following.

4.6.20.1 Standby power output performance test.

4.6.20.1.1 Performance at -26°C. Battery sample 2 shall be charged per 4.5.1.3 and then placed on open-circuit at -26°C for 12 ± .5 hours. The battery, while remaining in the -26°C chamber, shall be discharged into a 16-ohm resistive load for 1.0 hour. The battery shall maintain a terminal potential of 24.0 volts or greater during the discharge. The battery shall be removed from the chamber and immediately subjected to three start cycles per Figure 2, with a 2-minute rest period between starts. The battery shall meet the requirements of 3.6.23.

4.6.20.1.2 Performance at -18°C. Battery sample 3 shall be tested in accordance with 4.6.20.1.1, except that the test temperature shall be -18°C and the discharge time into 16 ohms shall be 2.0 hours.

4.6.20.1.3 Performance at 0°C. Battery sample 1 shall be tested in accordance with 4.6.20.1.1, except that the test temperature shall be 0°C and the discharge time into 16 ohms shall be 4.0 hours.

4.6.20.1.4 Performance at 25°C. Battery sample 2 shall be tested in accordance with 4.6.20.1.1, except that the test temperature shall be 25°C and the discharge time into 16 ohms shall be 7.0 hours.

4.6.20.1.5 Performance at 49°C. Battery sample 1 shall be tested in accordance with 4.6.20.1.1, except that the test temperature shall be 49°C and the discharge time into 16 ohms shall be 7.0 hours.

4.6.20.2 Cold and hot temperature charge test.

4.6.20.2.1 Half-hour charging. Battery sample 2 shall demonstrate the cold and hot temperature charging requirements of 3.6.24. The test shall be conducted as follows.

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a. Charge battery at 28.25 ± 0.5V for 2 hours at 25°C.

b. Stabilize the battery at -18°C, then discharge at 24.0A to 18.0V. The battery shall provide at least 12.0Ah capacity.

c. Stabilize the battery at 25°C, then charge at 28.25 ± 0.5V for 2 hours.

d. Stabilize the battery at -18°C, then discharge at 24.0A to remove all but 12.0Ah, as determined by 4.6.20.2.1b.

e. Rest battery to stabilize at test temperature, then apply 28.25 ± 0.5V for 30 minutes.

f. Immediately following charge, remove battery from test chamber and discharge at 24.0A to 18.0V. The battery must deliver 18.0Ah minimum.

4.6.20.2.1.1 The battery shall be tested in accordance with 4.6.20.2.1, except that the test temperature shall be 55°C.

4.6.20.2.2 One-hour charging. Battery sample 3 shall be subjected to a -40°C charge demonstration test and shall be conducted as follows:

a. Charge the battery at 28.25 ± 0.5V for 2 hours at 25°C.

b. Stabilize the battery at -40°C, and then discharge at 24.0A to 18.0V.

c. Rest battery to stabilize at -40°C, then apply 28.25 ± 0.5V for 1.0 hour.

d. Immediately following charge, remove battery from test chamber and discharge at 24.0A to 18.0V. Battery shall deliver 18.0Ah minimum.

4.6.23 Vibration. Delete and add the following: "Endurance and performance random test. The battery assembly, including connectors and cable, shall be mounted to the test apparatus, per 4.5.5, and subjected to the random vibration test procedure for equipment installed in jet aircraft, Procedure I.A, Method 514.2 of MIL-STD-810. The battery shall be vibrated along each orthogonal axis in accordance with the Random Vibration Test levels specified in Figure 4 of this specification sheet for a period of 21 minutes per axis. The battery heaters shall be energized and the battery charged at 28.25 ± 5 volts during vibration and shall be discharged at 156A for 1 minute prior to beginning each axis of vibration. The cables connected to the battery shall consist of four size AN-6 cables (two connected to the positive terminal and two connected to the negative terminal) and shall be clamped to the test apparatus at a cable length of 46 ± 3 centimeters from the battery."

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4.6.26 Ground storage. Line 4.6.26d, delete and add the following: "Subject the battery to three start cycles per Figure 2, with a 2-minute rest period between starts."

Preparing Activity:
Navy - AS
Project No. 6140-N661

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TABLE 1. Inspection of batteries and order of test.

Inspection Number	Inspections	Qualification Inspection: Qualification Sample Number 1	2	3	Requirements Paragraph	Method of Inspection Paragraph	Quality Conformance Inspection: Group A Tests	Group B Tests 1	2
1	Visual & Mechanical Examination	X	X	X	3.5, 3.6.2.2 & 3.6.3	4.6.3	X	X	X
2	Dimensions & Weight	X	X	X	3.6.5	4.6.4	X	X	
3	Dielectric Strength	X	X	X	3.6.4	4.6.5	X	X	X
4	Strength of Vent Tubes	X	X	X	3.6.7	4.6.7	X	X	X
5	Color and Marking	X	X	X	3.6.3	4.6.8	X	X	X
6	Conditioning Charge	X	X	X	4.5.1.4	4.6.9		X	X
7	Capacity Discharge	X	X	X	3.6.8	4.6.10	X	X	X
8	Constant Voltage Discharge	X	X		3.6.8	4.6.11	X	X	
9	Strength of Receptacle			X	3.6.9	4.6.12			
10	Standby Power Output Performance Test	X	X	X	3.6.23	4.6.20.1			
11	Cold & Hot Temperature Charge Test	X	X	X	3.6.24	4.6.20.2			
12	Life Cycling	X			3.6.10	4.6.13		X	
13	Capacity Discharge at 49°C		X		3.6.8	4.6.14			
14	Discharge While Inverted		X		3.5.8 & 3.6.8	4.6.16			X
15	Altitude		X		3.6.11 & 3.6.12	4.6.17			X
16	Mechanical Shock	X			3.6.11 & 3.6.12	4.6.18			
17	Temperature Shock	X			3.6.11 & 3.6.12	4.6.19		X	
18	Temperature Rise & Float		X		3.6.13	4.6.21			X
19	Battery Gas Emission Test		X		3.6.14	4.6.22			
20	Vibration		X		3.6.16	4.6.23			X
21	Humidity		X		3.6.17	4.6.24			X
22	Salt Fog	X			3.6.18	4.6.25		X	
23	Ground Storage			X	3.6.15	4.6.26			
24	Physical Integrity at High Temperature	X			3.6.12	4.6.27			
25	Final Examination	X	X	X	3.6.4	4.6.30		X	X

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